

FROM LOCAL RAPIDITY FIBER TO $\text{SPIN}(8)$: THE PRIMITIVE RAPIDITY OBSERVABLE AND 28-DIMENSIONAL CLOSURE

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ABSTRACT. We present a single derivation chain on a discrete index space. A uniform polynomial growth law forces a nontrivial cumulative curvature channel K . Normalizing this channel yields the primitive rapidity observable $\mathcal{R}$, which is sign-flip invariant and canonically forces exact 2π -periodic holonomy together with the index-growth reflection symmetry $[x, |x|]$. The geometric phase direction supplied by $\mathcal{R}$ produces the phase-lift generator Δ . Adjoining Δ to the 14 generators of G_2 then closes, in the Lie sense, to the full 28-dimensional algebra $\mathfrak{so}(8)$. The same construction generalizes to any $\text{SO}(n)$ via the Fourier-twiddle route on S^{n-1} . Historically this closure was first obtained through the full rapidity fiber; the concise $G_2 + \Delta$ route is its algebraic distillation. All nontrivial closure, transport, and holonomy steps are machine-checked in Lean 4.

1. INTRODUCTION

This paper presents one continuous geometric-to-algebraic derivation:

$$\text{uniform growth} \implies K \implies \Omega \implies \mathcal{R} \implies \Delta \implies \langle G_2, \Delta \rangle_{\text{Lie}} = \mathfrak{so}(8).$$

Historically, the closure was first reached through the full rapidity-fiber construction; the cleaner $G_2 + \Delta$ closure theorem used here is the compressed algebraic form of that original route.

Write *HQIV* for *Horizon-Quantized Informational Vacuum*: the framework that ties entanglement monogamy on overlapping causal horizons to discrete null-shell combinatorics and related readouts, as developed in the companion physics manuscript and machine-checked in the [1] library. Within HQIV, the chosen growth law is physics-motivated by discrete mode counting for a local light-cone construction on flat background space. The same analytic mechanism applies to any growth law that yields a positive divergent cumulative channel, including non-physics settings.

Let $m \in \mathbb{N}$ be a discrete index parameter and define

$$N(m) := (m+2)(m+1), \quad A(m) := 4N(m),$$

with fixed exponent

$$\alpha := \frac{3}{5}.$$

We explicitly credit this $\alpha = 3/5$ normalization to Brodie's thermodynamic horizon analysis [2]; the lattice-ratio identity is the formal verification anchor in the companion Lean development. For the analytic growth $\rightarrow$ curvature chain itself, any fixed $\alpha > 0$ preserves the positivity and divergence arguments below; $\alpha = 3/5$ is the distinguished normalization used in this paper. Define curvature density and cumulative curvature

$$\rho(x) := \frac{1}{x}(1 + \alpha \log x) \quad (x \geq 1), \quad K(n) := \sum_{m=0}^{n-1} \rho(m+1).$$

Fix any normalization index m_* with $K(m_*) > 0$ and set

$$\Omega(n) := \frac{K(n)}{K(m_*)}.$$

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Historical derivation arc. The development history matters for interpretation. The original route was

$$\text{rapidity fiber} \longrightarrow \text{phase-lift direction } \Delta \longrightarrow \mathfrak{so}(8) \text{ closure.}$$

In that route, the rapidity phase channel provides the geometric phase direction whose infinitesimal action is encoded by Δ . The present paper then uses a cleaner algebraic route: once Δ is isolated, adjoining it to the 14 generators of G_2 yields the full closure statement.

This should be read as a compression, not a replacement: the concise $G_2 + \Delta$ theorem is the distilled algebraic core of the earlier phase-fiber derivation. The formal claims in this manuscript concern the compressed route; the historical geometric motivation is included to keep the two parts of the story logically connected.

2. UNIFORM GROWTH AND THE FORCED CURVATURE CHANNEL

This is Step 1 of the derivation: growth law $\rightarrow \rho(x) \rightarrow K(n) \rightarrow \Omega(n)$.

Proposition 2.1 (Exact growth identities). *For all $m \in \mathbb{N}$,*

$$A(m) = 4m^2 + 12m + 8, \quad A(m+1) - A(m) = 8(m+2).$$

Proof. Expand $A(m) = 4(m+2)(m+1)$ and subtract successive values. $\square$

Remark 2.2 (Why this ρ and why $\alpha = \frac{3}{5}$). *Proposition 2.1 shows that the discrete derivative of the growth law is linear: $A(m+1) - A(m) = 8(m+2)$. A scale-invariant cumulative channel compatible with linear step growth is naturally of logarithmic-harmonic form*

$$\rho(x) = \frac{c + d \log x}{x},$$

whose partial sums produce harmonic and log-corrected terms. The normalization used in this paper is $c = 1$ and $d = \alpha = \frac{3}{5}$, following Brodie's horizon-thermodynamic coefficient choice [2], matching the exact lattice-ratio identity, and giving immediate positivity on $x \geq 1$. Within the HQIV framework this quadratic ladder is the discrete light-cone shell model; outside that framework it can be replaced by alternative growth laws while preserving the same proof pattern whenever positivity/divergence hypotheses are retained.

Lemma 2.3 (Positivity of density). *For all $x \geq 1$, $\rho(x) > 0$.*

Proof. Since $x \geq 1$, $\log x \geq 0$. Therefore $1 + \alpha \log x \geq 1 > 0$ and $1/x > 0$, so $\rho(x) > 0$. $\square$

Remark 2.4 (Parameter robustness). *The proofs of positivity of ρ , the lower bound $K(n) \geq H_n$, and therefore divergence of K , require only $\alpha > 0$ (indeed $\alpha \geq 0$ suffices for the lower bound). Thus the logical chain*

$$\text{growth} \rightarrow \rho \rightarrow K \rightarrow \Omega \rightarrow \mathcal{R}$$

is structurally robust under positive choices of α ; the value $3/5$ is a model-specific normalization rather than a necessity for the abstract analytic mechanism.

Theorem 2.5 (Uniform-growth curvature theorem). *For all $n \geq 1$,*

$$K(n) \geq H_n := \sum_{m=0}^{n-1} \frac{1}{m+1}.$$

Hence $K(n) \rightarrow +\infty$ as $n \rightarrow \infty$.

Proof. By definition,

$$\rho(m+1) = \frac{1}{m+1} (1 + \alpha \log(m+1)) \geq \frac{1}{m+1}$$

because $\log(m+1) \geq 0$. Combined with the linear step-growth from Proposition 2.1 and the logarithmic-harmonic ansatz in Remark 2.2, summing gives $K(n) \geq H_n$. Since H_n diverges, so does $K(n)$. $\square$

Remark 2.6. *The conclusion is pointwise and asymptotic: each finite partial sum is bounded below by harmonic growth, and the entire channel diverges.*

3. RAPIDITY AS A PRIMITIVE PHASE OBSERVABLE

This is Step 2: $\Omega \rightarrow \mathcal{R}$. In this manuscript, rapidity is treated as the first-order observable canonically forced by the normalized divergent channel Ω , and then used to define Δ .

Definition 3.1 (Normalized curvature readout). *Let $\Omega(n) := K(n)/K(m_*)$ with $K(m_*) > 0$.*

Lemma 3.2.

$$\Omega(m_*) = 1, \quad \Omega(n) \rightarrow +\infty \text{ as } n \rightarrow \infty.$$

Proof. The first identity is immediate from normalization. The second follows from Theorem 2.5 and positivity of $K(m_*)$. $\square$

Definition 3.3 (Rapidity phase channel). *Define rapidity as a normalized phase readout*

$$\mathcal{R}(\phi, t, n) := \mathcal{F}(\phi, t, \Omega(n)),$$

where $\mathcal{F}$ is the phase map. At $n = m_*$, this recovers the base harmonic phase.

Rapidity is treated here as a first-order primitive: the canonical phase observable forced directly by any divergent normalized curvature channel.

Remark 3.4 (Phase-map axioms used here). *For the derivations below, $\mathcal{F}$ is only required to satisfy:*

- *normalization at the reference index: $\mathcal{F}(\phi, t, 1) = \mathcal{H}(\phi, t)$;*
- *monotonic dependence on curvature channel: $\Omega_1 \leq \Omega_2 \Rightarrow \mathcal{F}(\phi, t, \Omega_1) \leq \mathcal{F}(\phi, t, \Omega_2)$ (for fixed (ϕ, t));*
- *embedding invariance: for any embedding e , pullback commutes with evaluation, i.e. $(\mathcal{F} \circ (\phi, t, \Omega)) \circ e = \mathcal{F} \circ (\phi \circ e, t \circ e, \Omega \circ e)$.*

No stronger closed form is needed for the closure and transport statements in this manuscript.

Remark 3.5 (Concrete phase-map instance). *To avoid black-box ambiguity, one admissible concrete choice is*

$$\mathcal{F}(\phi, t, \Omega) := \Omega \cdot \mathcal{H}(\phi, t),$$

with $\mathcal{H}$ the base harmonic phase. This satisfies normalization $\mathcal{F}(\phi, t, 1) = \mathcal{H}(\phi, t)$, is monotone in Ω whenever $\mathcal{H}(\phi, t) \geq 0$, and realizes the same discrete chain $\Omega(n) \mapsto \mathcal{R}(\phi, t, n)$.

Remark 3.6 (Interpretive bridge to classical rapidity and geometric phase). *The object $\mathcal{R}(\phi, t, n)$ is discrete in this manuscript, but it plays the same structural role as a hyperbolic-angle parameter (classical relativistic rapidity) in that it indexes phase/boost-like updates along a one-parameter channel. The phase-direction language is also consistent with geometric-phase viewpoints (Berry and Aharonov–Anandan) at the level of interpretation; no additional continuum assumptions are required for the proofs here.*

Remark 3.7. *The construction remains explicitly chained*

$$\text{growth} \rightarrow K \rightarrow \Omega \rightarrow \mathcal{R},$$

but the role of $\mathcal{R}$ is primitive at the level of output type: it is the canonical first-order phase observable attached to any divergent normalized channel in this framework.

Sign-flip invariance. The rapidity channel is invariant under simultaneous sign reversal of the raw curvature channel and phase increment data:

$$\rho \mapsto -\rho, \quad K \mapsto -K, \quad \Omega \mapsto \pm\Omega, \quad \theta \mapsto -\theta.$$

At generator level, $\Delta \mapsto -\Delta$ changes orientation of the same distinguished plane action and preserves 2π -periodic closure. This invariance guarantees that the forced 2π -periodic holonomy and the reflection symmetry $[x, |x|]$ survive under reversal of the curvature channel.

Universal outputs in the present framework. Given a divergent cumulative channel and the phase-lift construction of Section 4, the framework-level outputs are:

- exact 2π -periodic plane holonomy generated by Δ ;
- reflection pairing symmetry in index-growth form ($[x, |x|]$ / reciprocal pairing language);
- full Lie-closure to $\mathfrak{so}(8)$ (or more generally to $\mathfrak{so}(n)$) once a suitable seed subalgebra is adjoined to the phase-lift generator (see Theorem 3.9 and Section 5).

Remark 3.8 (Discrete channel and “fiber” language). *The core proofs in this paper are discrete ($n \in \mathbb{N}$). The term “rapidity fiber” is used in the minimal sense of a one-parameter phase direction attached to each index state through $\mathcal{R}(\phi, t, n)$, not as a fully developed smooth bundle construction. A continuum/fiber-bundle realization is treated as an interpretation layer motivated by embedding invariance and thermodynamic-limit heuristics, not as an additional theorem used in Sections 2–5.*

Optional abstract transport interface. The following transport picture is an optional abstract interface and is not used in the core discrete proofs. Let X be a carrier set (or space) with maps $e_1 : X_1 \rightarrow X$ and $e_2 : X_2 \rightarrow X$ and a shared observable $r : X \rightarrow \mathbb{R}$. Then

$$r_1 := r \circ e_1, \quad r_2 := r \circ e_2$$

are induced channels on each domain. Any bound on r over one image transports to the other channel under the same threshold hypotheses.

$$\begin{array}{ccccc} X_1 & \xrightarrow{e_1} & X & \xleftarrow{e_2} & X_2 \\ r_1 \downarrow & & r \downarrow & & \downarrow r_2 \\ \mathbb{R} & \xlongequal{\quad} & \mathbb{R} & \xlongequal{\quad} & \mathbb{R} \end{array}$$

Hence a bound $r \leq \tau$ on one embedded image is transported through the same ambient observable.

Generalization to $\mathrm{SO}(n)$ via Fourier twiddles. The phase-lift mechanism is dimension-independent once a distinguished 2-plane is fixed. For $n \geq 2$, choose $i < j$ and let $\Delta_{ij} \in \mathfrak{so}(n)$ be the standard antisymmetric plane-rotation generator (nonzero 2×2 block on $\mathrm{span}\{e_i, e_j\}$, zero elsewhere). For rapidity increment $\theta(n)$, define

$$U(\varepsilon, n) := \exp(\varepsilon \theta(n) \Delta_{ij}) \in \mathrm{SO}(n).$$

On $L^2(S^{n-1})$, the induced weight- k multiplier is the twiddle character

$$\chi_k(\theta(n)) = \exp(i k \theta(n)), \quad k \in \mathbb{Z}.$$

Theorem 3.9 (Universal phase-lift holonomy equation). *For every $n \geq 2$ and every embedded plane generator Δ_{ij} ,*

$$\exp(2\pi \Delta_{ij}) = I_n,$$

equivalently

$$\mathrm{tr}(\exp(t \Delta_{ij})) = n - 2 + 2 \cos t,$$

so $\mathrm{tr}(\exp(t \Delta_{ij})) = n$ whenever $t = 2\pi k$, $k \in \mathbb{Z}$.

Proof. Δ_{ij} is conjugate inside $\mathfrak{so}(n)$ to the block matrix with $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ on one 2-plane and zeros elsewhere. Exponentiation gives one planar rotation block $\begin{pmatrix} \cos t & \sin t \\ -\sin t & \cos t \end{pmatrix}$ and identity on the orthogonal complement, yielding the trace formula. At $t = 2\pi k$, the rotation block is identity, so the full matrix is I_n . The trace identity recovers the classical 2π -periodicity of rotations in any embedded plane. $\square$

Remark 3.10. *The $\mathfrak{so}(8)$ case is distinguished by the octonionic seed choice $(G_2 \cup \{\Delta\})$; the twiddle/holonomy mechanism itself is canonical for all $n \geq 2$ in this framework.*

4. THE PHASE-LIFT GENERATOR Δ

This is the bridge step from geometry to algebra. In the historical route, rapidity is first constructed as a phase observable $\mathcal{R}$, and its infinitesimal phase direction is represented by the anti-symmetric generator Δ . The concise route used for closure starts directly from this same Δ .

Explicit construction from the rapidity channel. Fix a reference index m_* and define the rapidity phase increment

$$\theta(n) := \mathcal{R}(\phi, t, n) - \mathcal{R}(\phi, t, m_*).$$

On the distinguished phase plane $P = \text{span}\{e_1, e_7\} \subset \mathbb{R}^8$, define the one-parameter action

$$U(\varepsilon, n) := \exp(\varepsilon \theta(n) \Delta), \quad U(0, n) = I_8.$$

The infinitesimal generator is

$$\left. \frac{d}{d\varepsilon} U(\varepsilon, n) \right|_{\varepsilon=0} = \theta(n) \Delta.$$

In the standard basis, the phase-lift generator used in the closure seed is the skew-adjoint matrix with only two nonzero entries:

$$\Delta_{1,7} = 1, \quad \Delta_{7,1} = -1, \quad \Delta_{ij} = 0 \text{ otherwise,}$$

equivalently the 2×2 rotation block $\begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix}$ on P and zero on $P^\perp$. Thus $\Delta \in \mathfrak{so}(8)$ is the infinitesimal phase-direction lift of the rapidity channel.

Remark 4.1 (Why the e_1 - e_7 plane). *The e_1 - e_7 phase plane is the canonical choice used by `HQVM/matrices.py`: e_7 is the color-preferred octonionic axis in the fixed Fano-basis realization, and Δ is defined as the infinitesimal rotation mixing e_1 with e_7 . This aligns the geometric phase lift with the same basis conventions used by the G_2 generator stack and triality-aware embedding data. In the same multiplication-table convention, e_7 is the G_2 -stabilized color axis while e_1 is the orthogonal phase/electromagnetic direction, so the e_1 - e_7 plane is the minimal triality-compatible phase-lift plane in this basis.*

Remark 4.2 (Algebraic role of Δ). *Beyond antisymmetry, the closure mechanism requires that Δ contributes a direction not exhausted by the raw linear span of G_2 . This is certified in the formal development by `finrank_span_G2_union_Delta_le_15` together with `exists_so8Generator_not_mem_span_G2_union_Delta`, and completed by `lieBracket_in_span` and `G2_plus_Delta_closes_to_so8`.*

Proposition 4.3 (Generator-level equivalence bridge). *The historical geometric route and the concise algebraic route are equivalent at generator level:*

$$(\text{rapidity fiber} \Rightarrow \Delta) \iff (\Delta \text{ used as the algebraic seed in } G_2 + \Delta).$$

Proof. Both routes identify the same phase-lift direction Δ . The geometric route derives it from the rapidity phase channel; the algebraic route takes that identified Δ as input to Lie closure. Thus they differ by presentation length, not by generator content. $\square$

5. ALGEBRAIC CLOSURE: $G_2 + \Delta \rightarrow \mathfrak{so}(8)$

This is Step 3 and the algebraic centerpiece: the distilled closure theorem.

5.1. Algebraic frame. Let $\mathfrak{g}_2$ denote the 14-generator octonionic derivation sector and Δ the phase-lift antisymmetric generator in $M_8(\mathbb{R})$. Let $\{X_k\}_{k=1}^{28}$ denote the explicit antisymmetric basis candidate.

Lemma 5.1 (Ambient model). *The space of real skew-adjoint 8×8 matrices has dimension 28:*

$$\dim\{M \in M_8(\mathbb{R}) : M^\top = -M\} = 28.$$

Proof. A skew-adjoint matrix is determined by strict upper-triangle entries; there are $\binom{8}{2} = 28$ such coordinates. $\square$

Lemma 5.2 (Seed containment). *Every generator in $\mathfrak{g}_2$ and Δ lies in the skew-adjoint model, hence in $\mathfrak{so}(8)$.*

Proof. Each seed satisfies $M + M^\top = 0$ by construction. $\square$

Lemma 5.3 (Linear-span obstruction). *The linear span of the raw seed set $G_2 \cup \{\Delta\}$ has dimension at most 15, so it cannot equal $\mathfrak{so}(8)$.*

Proof. At most 15 listed matrices are linearly combined directly; but $\dim \mathfrak{so}(8) = 28$. $\square$

Theorem 5.4 (Full Lie closure). *The Lie algebra generated by $G_2 \cup \{\Delta\}$ is all of $\mathfrak{so}(8)$.*

Proof. Let $\mathfrak{h} := \langle G_2, \Delta \rangle_{\text{Lie}}$.

Step 1. The candidate family $\{X_k\}_{k=1}^{28}$ is linearly independent and closed under commutator:

$$[X_i, X_j] = \sum_{k=1}^{28} c_{ij}^k X_k.$$

Hence $\text{span}\{X_k\}$ is a 28-dimensional Lie subalgebra of skew-adjoint matrices.

Step 2. Iterated commutators from $G_2 \cup \{\Delta\}$ generate all 28 basis directions (closure certificate), so $\dim \mathfrak{h} = 28$.

Step 3. Since $\mathfrak{h} \subseteq \mathfrak{so}(8)$ and both have dimension 28, equality follows:

$$\mathfrak{h} = \mathfrak{so}(8).$$

$\square$

Remark 5.5. *The key distinction is structural: the linear seed span is not full, but the Lie-generated span is full.*

Toy model: quaternionic closure $\mathfrak{so}(3) + \Delta_4 \rightarrow \mathfrak{so}(4)$. The same linear-vs-Lie mechanism appears in the smallest nontrivial setting. Let

$$J_{12}, J_{13}, J_{23} \in \mathfrak{so}(4)$$

be the standard $\mathfrak{so}(3)$ seed acting on the first three coordinates, and let

$$\Delta_4 := J_{14}$$

be the phase-lift generator (direct analog of the distinguished phase-plane lift in Section 4). Then:

- the raw linear span of $\{J_{12}, J_{13}, J_{23}, \Delta_4\}$ has dimension $4 < 6 = \dim \mathfrak{so}(4)$;
- iterated brackets generate the missing directions J_{24}, J_{34} , since

$$[J_{12}, J_{14}] = J_{24}, \quad [J_{13}, J_{14}] = J_{34};$$

- therefore $\langle \mathfrak{so}(3), \Delta_4 \rangle_{\text{Lie}} = \mathfrak{so}(4)$.

So the closure pattern is already visible in a hand-checkable 6-dimensional case before the octonionic 28-dimensional setting (the same linear-vs-Lie mechanism and 2π -holonomy hold verbatim for any $\mathrm{SO}(n)$).

For reproducibility, the same exact-rational symbolic pipeline used for the $\mathfrak{so}(8)$ certificate is mirrored in a toy script (`scripts/generate_symbolic_so4_certificate.py`) writing `artifacts/so4_symbolic_certificate.json`.

Explicit bracket strategy (illustrative fragment). The closure certificate is built by working in a fixed skew-adjoint basis $\{X_k\}_{k=1}^{28}$ and computing all commutators in coordinates:

$$[X_i, X_j] = \sum_{k=1}^{28} c_{ij}^k X_k.$$

The computational pipeline constructs the 28×28 basis matrix, inverts it once, and then projects each bracket vector to recover the coefficients c_{ij}^k . Lean then checks the resulting coefficient table inside the formal algebra.

A representative fully explicit row (from the generated closure table) is:

$$[X_0, X_1] = \sum_{k=0}^{27} c_{01}^k X_k,$$

with sample coefficients

$$c_{01}^2 = \frac{-13165551793093}{100000000000000}, \quad c_{01}^{18} = \frac{-17776643565617}{200000000000000}, \quad c_{01}^{27} = \frac{-1315768990787}{250000000000000},$$

and exact reconstruction identity in the symbolic certificate

$$[X_0, X_1] - \sum_k c_{01}^k X_k = 0.$$

This demonstrates the certificate pattern used for all pairs (i, j) .

For searchability in rendered PDFs, we also record the exact textual excerpt emitted by the generator script:

```
| 0, 1, 2 => ((-13165551793093 : R) / 100000000000000)
| 0, 1, 18 => ((-17776643565617 : R) / 200000000000000)
| 0, 1, 27 => ((-1315768990787 : R) / 250000000000000)
-- Verification [0,1]:
max |bracket - sum c_k gen_k| = 2.220446049250313e-16
```

These lines are produced by `scripts/print_lie_bracket_closure.py` (`main()`, bracket check near the final verification print block). Reproducibility note: this Python script is a coefficient generator (not itself formally verified). Formal verification occurs in Lean by importing the emitted tables (`GeneratorsLieClosureData0–GeneratorsLieClosureData27`, assembled by `GeneratorsLieClosureData.lean`) and proving `lieBracket_in_span` and `G2_plus_Delta_closes_to_so8`. Thus the certificate is machine-checked at theorem level even though coefficient production is computational. In parallel, we now provide an exact-arithmetic symbolic proof-object generated by `scripts/generate_symbolic_lie_closure_certificate.py`, exported as `artifacts/so8_symbolic_certificate.json`. That artifact records a rank-28 basis and all structure coefficients over $\mathbb{Q}$ (numerator/denominator pairs), with exact linear solves for each bracket relation.

Remark 5.6 (Methodological caveat: numerical certificate vs exact arithmetic). *The closure package now contains two complementary certificates: (i) the original floating-point Lean-consumed table with machine-epsilon residual checks, and (ii) an exact symbolic certificate over $\mathbb{Q}$. The first is directly wired into the current Lean closure modules; the second is an exact proof-object for strict audit trails.*

For the “outside-seed-direction” mechanism, one may choose a witness $X_{k_\star}$ outside the raw seed span (guaranteed by the linear obstruction theorem) and then produce a short Lie word $W_\star$ in $G_2 \cup \{\Delta\}$ such that

$$W_\star = \sum_{k=1}^{28} a_k X_k, \quad a_{k_\star} \neq 0,$$

typically appearing after two or three nested brackets in the closure search.

To bridge linear span ≤ 15 to full Lie closure = 28, the proof uses:

- (1) linear obstruction: `finrank_span_G2_union_Delta_le_15`;
- (2) basis directions outside raw seed span: `exists_so8Generator_not_mem_span_G2_union_Delta`;
- (3) exhaustive commutator closure and dimension witness: `lieBracket_in_span` and `G2_plus_Delta_closes_to_so8`.

So the additional directions are not assumed: they are produced by iterated bracketing and certified by the full coefficient table.

6. METHODOLOGICAL UNIFICATION

Proposition 6.1 (Shared forced-structure pattern). *Assume:*

- (1) a uniform polynomial growth law on a discrete index space, with $\alpha > 0$, producing the channel K as in Theorem 2.5; and
- (2) an octonionic seed $G_2 \cup \{\Delta\} \subset \mathfrak{so}(8)$ with the certified Lie-closure data from Section 5.

Then three universal outputs are forced from minimal input data:

- a sign-flip-invariant rapidity observable $\mathcal{R}$;
- exact 2π -periodic holonomy on the distinguished phase plane;
- upon adjoining a suitable seed algebra, full Lie closure to $\mathfrak{so}(n)$ (specialized in this paper to $\mathfrak{so}(8)$).

Proof. The first conclusion is exactly the content of Theorem 2.5, the normalization lemma for Ω , and Definition 3.3 of the phase readout $\mathcal{R}$. The second conclusion is Theorem 5.4 together with the linear-vs-Lie distinction made explicit by the obstruction lemma. All three are “forced-structure” statements: divergence and positivity force a nontrivial rapidity observable; the phase-lift action forces exact planar 2π -periodicity; and iterated commutators force completion to the full 28-dimensional algebra in the octonionic seed case. $\square$

This proposition is methodological, not causal: it does not assert a physical derivation of $\mathfrak{so}(8)$ from the growth channel (or vice versa), only that both sections instantiate the same minimal-generator $\Rightarrow$ rigid-closure proof pattern. In particular, the algebraic closure theorem is logically independent of the growth parameters once Δ is specified; the growth channel supplies a geometrically natural route to that seed through the rapidity construction.

7. FOUNDATIONAL AND ADJACENT LITERATURE

This paper touches four classical lines of work.

Octonions, G_2 , triality, and $\mathfrak{so}(8)$. The octonionic derivation algebra G_2 and the triality structure of $\text{Spin}(8)$ are standard foundations for exceptional and rank-8 geometry. Classical references include Cartan’s work on exceptional Lie groups and modern expositions such as Baez’s survey on octonions [3] and Springer–Veldkamp [9]. The present contribution is not a classification result, but a closure pipeline with an explicit *linear-vs-Lie* distinction and machine-checked closure certificate. On the physics side, $\text{SO}(8)$ /triality is foundational in superstring constructions and related compactification arguments [6]. More recent algebraic unification programs using division algebras

and octonionic embeddings, including Furey-style Standard Model embeddings, provide adjacent motivation for treating G_2 -to- $\mathfrak{so}(8)$ closure as physically meaningful structure [7]. This derivation also aligns with discretization programs emphasizing emergence of exceptional structure from combinatorial data (spin-foam/causal-set viewpoints), where rapidity acts as a first-order bridge from growth channels to orthogonal-group symmetry organization [3].

Growth-implies-structure paradigm. The statement that growth laws force geometric or algebraic structure is reminiscent of Gromov’s polynomial-growth rigidity paradigm [5]. In discrete and combinatorial geometry, curvature notions of Forman and Ollivier [4, 8] similarly connect local transport and growth behavior to global geometric constraints. Our curvature channel is a direct constructive instance of this philosophy: the chosen growth law yields an explicit positive, divergent cumulative curvature functional.

Rapidity and geometric phase viewpoints. Rapidity is classically a hyperbolic parameter in Lorentzian kinematics; geometric phase frameworks generalize phase observables to broader manifold settings. The novelty here is the derivation chain

$$\text{growth} \rightarrow K \rightarrow \Omega \rightarrow \mathcal{R},$$

which promotes rapidity from a coordinate choice to an induced index-space observable attached to a normalized curvature channel.

Formal verification context. Lean/mathlib is increasingly used for algebra and geometry formalization. This manuscript adopts a derivation-first style in the main text and records the nontrivial formal artifacts in Appendix A, so claims can be audited at proof-object level without turning the paper into a code index. For submission hygiene, all cited software artifacts (repository URLs, scripts, and generated certificates) should be evaluated at stable public snapshots/DOIs so referees can reproduce the exact reviewed state.

8. CONCLUSION

The derivation chain established in this paper is:

$$\text{uniform growth} \implies K \implies \Omega \implies \mathcal{R} \implies \Delta \implies \mathfrak{so}(8).$$

Historically, the long geometric route and the concise $G_2 + \Delta$ route are the same construction seen at two resolutions: the latter is the algebraic distillation of the former. What is proved here is the full derivation flow in a compact form, with closure certificates and transport components audited in Lean. Interpretively, the derivation chain is used here as a physics-motivated structural map (light-cone/discrete-shell setting), not as a claim that growth data alone physically causes $\mathfrak{so}(8)$. The same forced-structure template can be transplanted to other domains where cumulative growth observables are central. Although the primary result concerns the octonionic case, the same mechanism appears in a direct 6-dimensional analog in $\mathfrak{so}(4)$: scalar channel $\rightarrow$ phase-lift seed $\rightarrow$ Lie closure. The identical symbolic certificate pipeline provides an exact rational structure-constant table for the quaternionic seed ($\mathfrak{so}(3) + \Delta_4$), and the corresponding artifacts are included in the supplementary Lean bundle.

9. OUTLOOK (CONJECTURAL EXTENSIONS)

The following items are intentionally listed as conjectural/programmatic and are not proved in this manuscript:

- extension of the reflection-pairing mechanism to full critical-line constraints for zeta-type zero sets;
- systematic acceleration schemes for constants (π , odd zeta values) from the same twiddle/closure certificate stack in higher dimension;

- broader geometric interpretations (including holographic-style dual descriptions) driven by the same growth-to-holonomy template.

These are deferred to follow-up work; the present paper establishes only the proved core chain and its machine-checked closure/certificate components.

Schematic chain.

$$\text{growth} \implies K \implies \Omega \implies \mathcal{R} \implies \Delta \implies \text{Lie words} \implies \mathfrak{so}(n) \text{ closure},$$

generating downstream exact certificates for π , accelerated series for odd zeta values, and hypercomplex Fourier twiddles on S^{n-1} .

The same first-order rapidity construction retrofits classical rapidity on hyperbolic surfaces, geometric/Berry phases, and critical-line symmetry formulations for zeta-type functions through the index-growth reflection pairing $[x, |x|]$. It likewise provides a canonical bridge to hypercomplex Fourier analysis on S^{n-1} and matrix continued-fraction acceleration patterns for constants such as $\zeta(3)$ and π .

Remark 9.1 (Gravity-like information). *The forced 2π -periodic holonomy of the rapidity field is the discrete analog of curvature (parallel-transport deficit around closed loops). For evolving manifolds whose volume or mode-counting produces a divergent channel, the same primitive rapidity carries emergent geometric invariants and supplies a proto-holographic bridge from boundary growth data to bulk-like internal gauge symmetry $\mathfrak{so}(n)$.*

APPENDIX A. FORMAL VERIFICATION MAP

The text above is derivation-first. Corresponding machine-checked artifacts are:

Growth and curvature.

- `alpha_eq_3_5, latticeAlphaRatio_eq_alpha`
- `curvature_integral_ge_harmonic`
- `curvature_integral_tends_to_atTop`
- `omega_k_partial_at_reference`

Rapidity channel and transport interface.

- `rapidity_tip_recovers_harmonic_step`
- `shared_manifold_induces_common_rapidity`
- `shares_threshold_of_pointwise_bounds`
- `shares_threshold_of_step_bounds`

$\mathfrak{so}(8)$ closure.

- `so8_generators_linear_independent`
- `lieBracket_in_span`
- `span_so8Generators_eq_skewAdjointOne`
- `finrank_span_G2_union_Delta_le_15`
- `exists_so8Generator_not_mem_span_G2_union_Delta`
- `G2_plus_Delta_closes_to_so8`
- symbolic certificate generator: `scripts/generate_symbolic_lie_closure_certificate.py`
- exact proof-object artifact: `artifacts/so8_symbolic_certificate.json`
- dimension-general generator note: `scripts/generate_symbolic_lie_closure_certificate.py` accepts arbitrary n with embedded plane choice (i, j) , emitting corresponding Δ_{ij} and rational structure-constant data.
- toy-model symbolic generator: `scripts/generate_symbolic_so4_certificate.py`
- toy-model exact artifact: `artifacts/so4_symbolic_certificate.json`

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